Fluid Pressure Review

1. Pressure: $\boxed{p = \gamma \cdot h}$

SI: $\rho = 1\ Mg/m^3 = 1000\ kg/m^3$

$\gamma = 9.81 \cdot \rho = \boxed{9810\ N/m^3 = \gamma}$

US: $\boxed{\gamma = 62.4\ lb/ft^3}$

For fresh water. If fluid has $SG \neq 1$, then use $\gamma = \gamma_{fresh} \cdot SG$

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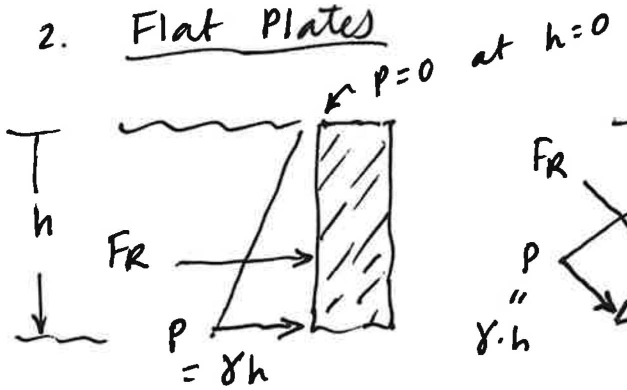
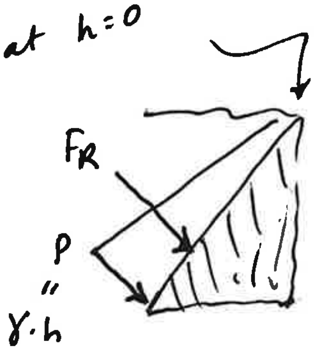
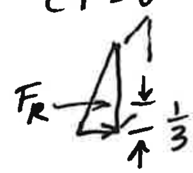
2. **Flat Plates**





(p = 0 at surface)



Triangular Pressure Distrib ⊥ to plate.

$\boxed{F_R = \frac{1}{2} b h \cdot (\text{width of plate})}$

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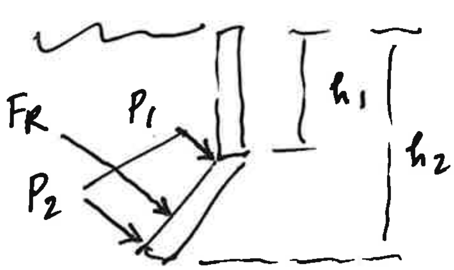
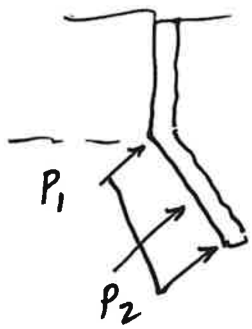
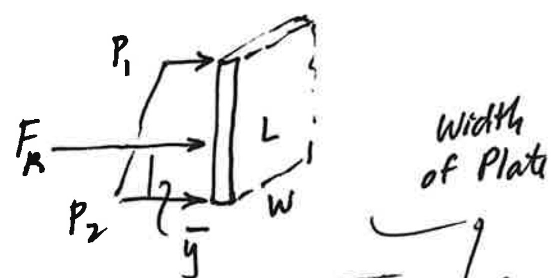
3. **Flat Plates (at depth)**





$$\boxed{\begin{aligned} P_1 &= h_1 \cdot \gamma \\ P_2 &= h_2 \cdot \gamma \end{aligned}}$$

Trapezoid



$$\boxed{F_R = \left(\frac{P_1 + P_2}{2}\right)(L)(W)}$$

$$\boxed{\bar{y} = \frac{1}{3}\left[\frac{2P_1 + P_2}{P_1 + P_2}\right] L}$$

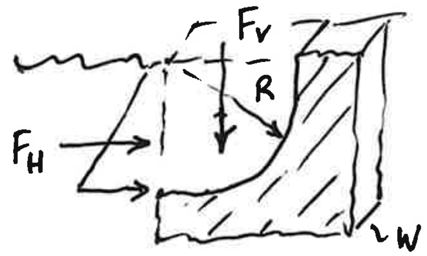
Fluid Pressure (cont'd)

4. **Curved Surfaces:** Cannot use a simple triangular or trapez pressure distribution.

Must use: $\boxed{\vec{F}_R = \vec{F}_H + \vec{F}_V}$

Examples:

1/4 Circle



Δ for F_H

Wt of fluid for F_V

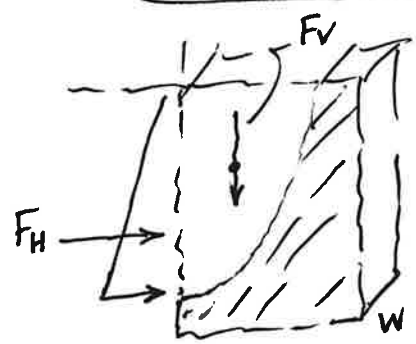
$Wt = VOL \cdot \gamma = F_V$

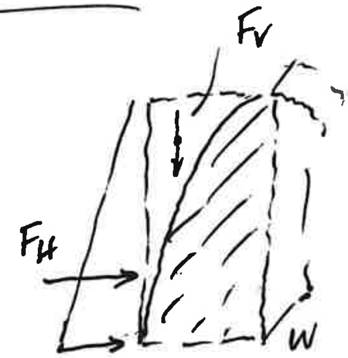
$$F_V = \underbrace{\left(\frac{\pi}{4} R^2\right)}_{\text{Area}}(W) \cdot \gamma$$

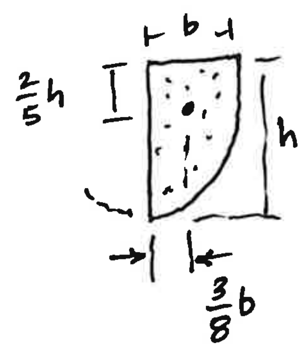
Acts at Centroid:

$$\bar{x} = \frac{4R}{3\pi}$$

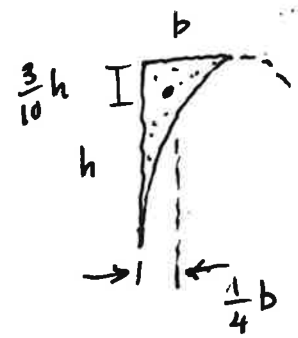
Parabola





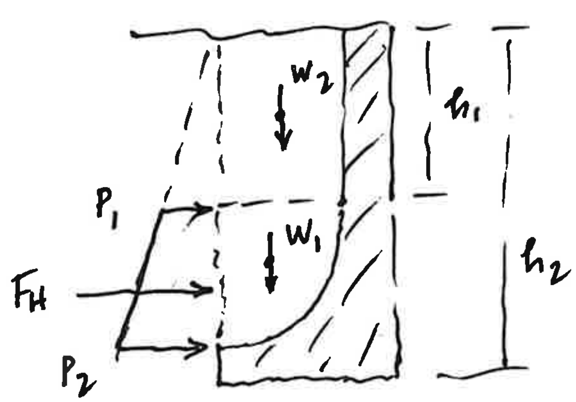


$A = \frac{2}{3}bh$



$A = \frac{1}{3}bh$

Curved Surface below water level...



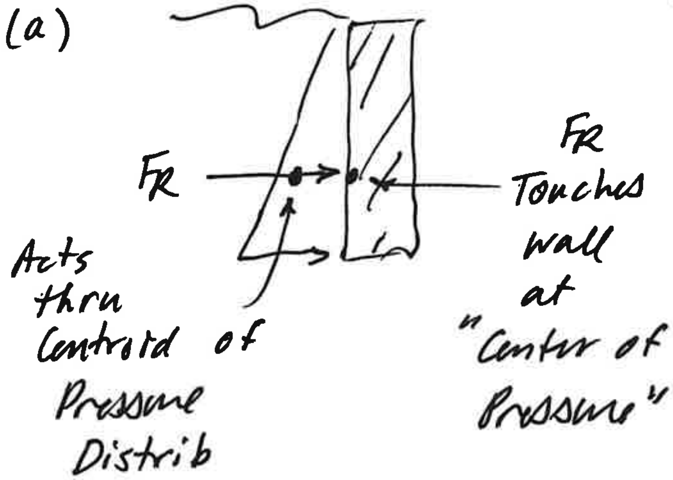
Trapezoid

$F_V = W_1 + W_2$

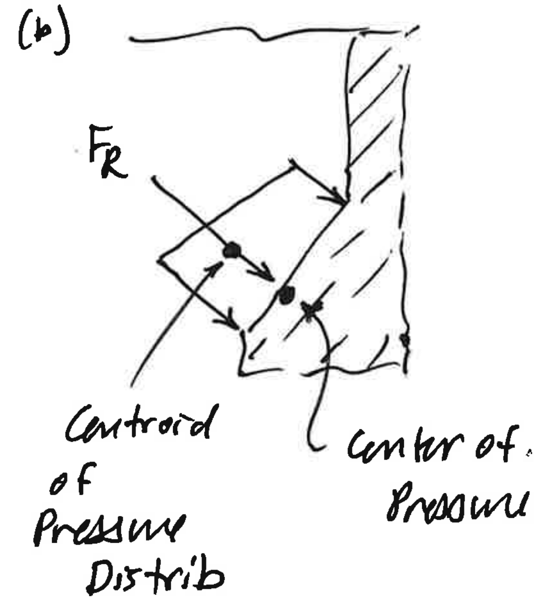
F_H = Trapezoid

5. "Center of Pressure": Point at which $\vec{F}_R$ touches wall or surface.

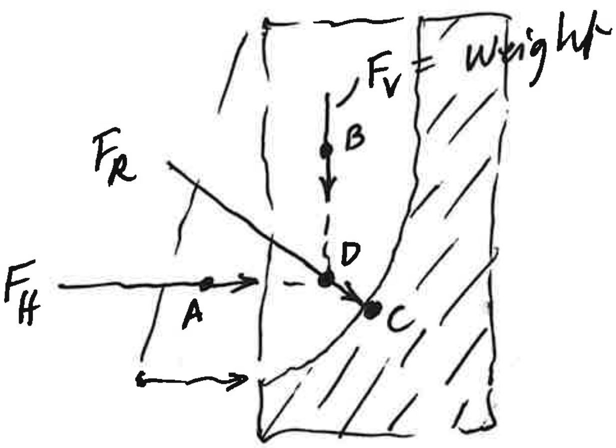
(a)



(b)



(c) Curved Surface



$$\vec{F}_R = \vec{F}_H + \vec{F}_V$$

A: Centroid of Δ
F_H acts thru A

B: Centroid of Weight
F_V acts thru B

D: Line of action of
F_R acts thru D..

$\subseteq$ "Center of Pressure": where $\vec{F}_R$ touches wall..